Why Opaque Refs Beat Brittle Selectors

Anti-Staleness Verification, Form Sanitization & Deterministic Element Resolution

Deterministic Element Targeting: Eliminating Broken Clicks

In modern web portals, asynchronous JavaScript frequently alters DOM elements between agent reasoning steps. Fathom implements an active **Anti-Staleness Guard** to guarantee that actions are executed only on valid, intended targets.

1. PRE-EXECUTION FRESHNESS CHECK

Before executing a `computer_click` or `computer_type` command, the runtime takes a microsecond snapshot to confirm that the target element ref (@e14) still exists and matches its original role fingerprint.

2. STALE-REF REJECTION & SELF-RECOVERY

If the page navigated or a popup closed during agent thinking, the ref is rejected immediately with an explicit error: *"Ref @e14 is stale. Capturing fresh snapshot."* The agent re-evaluates the new DOM state without misclicking.

Form Data Security & Workspace Confinement

Automatic Password Scrubbing

Password inputs and sensitive token fields are automatically masked in accessibility snapshots, preventing sensitive credentials from appearing in reasoning transcripts.

Confined File Workspace

Browser downloads are strictly confined to an isolated directory (`/data/browser`), preventing malicious downloads from touching host system files.

DETERMINISTIC ACTION SAFETY SCORECARD

100%	< 1ms	Zero	Masked
STALE REF DETECTION	TARGET VALIDATION	PHANTOM CLICKS	PASSWORD FIELDS

Zero Phantom Interactions: Fathom ensures that automated browser actions are as predictable, deterministic, and safe as compiled software code.